

Jason Carlson

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SUMMARY OF QUALIFICATIONS

AI Research Engineer with 5 years of experience building and deploying large-scale ML and RL systems (1B+ inferences/day, \$43M+ annual impact). Experience spans supervised learning, RL, and LLM fine-tuning, including clinical ML and human-aligned policy learning.

WORK EXPERIENCE

Independent AI Researcher

New York, NY

April 2025 – Present

- Developing methods to evaluate safety and fidelity of LLM-generated reasoning in clinical decision-making (aneurysm rupture risk), including analysis of misleading explanations and overconfidence
- Training preference-optimized policies (DPO variants) to better model human decision-making in chess ([garrychess.ai](#)). Accepted to IEEE CoG oral: [Paper Link](#)
- Conducting research on robust exploration methods in reinforcement learning (paper in preparation)

Amazon

New York, NY

Software Engineer II

May 2025 – April 2026

- Led development and deployment of a supervised credit-extension model trained on 1M+ payment-failure labels, replacing a rule-based system and generating \$43M annual profit (+200%)
- Designed RL-based credit allocation experiments (PPO + safety constraints) to optimize long-term outcomes under risk limits
- Fine-tuned a foundation model for ticket root-cause analysis using SFT + Ray (production pipeline)
- Stack: PyTorch, Ray, JAX, AWS

AppLovin

New York, NY

Senior Software Engineer

October 2024 – March 2025

- Built a large-scale identity graph system, reducing infrastructure cost by \$1.8M/year and improving cold-start latency by 87.5%
- Stack: Spark, Airflow, Java, Delta Lake, GCP

Amazon

New York, NY

Software Engineer I to II

June 2021 – September 2024

- Designed ML data pipeline supporting VP-level initiative; onboarded 5 engineers and productionized end-to-end workflow
- Built inference service for BERT-class embedding model generating 1B+ embeddings/day, deployed on AWS Inferentia
- Tech: Java, Python, Spark, AWS

Axosoft

Scottsdale, AZ

Software Developer Intern

May 2019 – July 2019

- Created Release Notes Tab & HTML file loader used by over a million software developers; used React.js, Redux, and Javascript

PUBLICATIONS

- Beyond Imitation: Preference-Optimized Policies for Realistic Grandmaster Chess—IEEE CoG 2026 oral. First author in a team of three. [Paper Link](#)
- The Use of AutoML for Predicting Intracranial Aneurysm Rupture... — Poster, CNS 2025. (Co-author; AutoML vs hand-engineered pipelines). [Paper Link](#)
- TopoExplore: Topology-Guided Exploration for RL — Research in progress. Benchmark suite + baselines (MiniGrid DoorKey); algorithm + ablations in progress

SKILLS

ML: Supervised learning, RL (PPO, constrained/safe RL), LLM fine-tuning, evaluation

Systems: Spark, Ray, large-scale ETL, distributed inference, vector search

Languages: Python, Java, C++, SQL

Infra: AWS, GCP, Docker, Kubernetes, Airflow

EDUCATION

Arizona State University

Tempe, AZ

Master of Arts: Mathematics (GPA: 4.0)

2021

Bachelor of Science: Computer Science (GPA: 3.9)

2020

Bachelor of Science: Mathematics

- *Awards:* Dean's Scholarship Recipient; Outstanding Junior: Math Department
- *Relevant Coursework:* Real Analysis II (A+), Geometry & Topology of Manifolds II (A), Functional Analysis II (A), Abstract Algebra II (A+), Probability (A), Data Structures & Algorithms (A+), Intro to AI (A)

PROJECTS: Hyperlinked to results and details

- [Chaos-Theory Website](#) (2020): Wrote simulations of Chaotic differential equations using React, Matlab, React-vis and displayed them on a website along with a bunch of theorems and relevant definitions
- [Search Algorithm Visualizer](#) (2020): Created a website that shows search algorithms visually. It includes DFS, BFS, and A* and uses React for the front-end
- **Berkley Pac-Man Projects** (2019): Wrote algorithms to optimize Pac-Man & Ghost AI movement choices; Earned 100% on all projects